

What is Agentic AI?

English Notes

Agentic AI using LangGraph (2026) — Playlist Notes

1 Definition

Agentic AI is a type of AI that can:

- Take up a task and goal from a user
- Work towards completing it on its own with minimal human guidance
- Plan, take actions, adapt to changes, and seek help only when necessary

2 Key Distinction

Aspect	Generative AI (ChatGPT)	Agentic AI
Interaction	Reactive — responds only to direct queries	Proactive — takes initiative autonomously
Initiative	Human must guide each step	System plans & executes on its own
Example	You ask step-by-step about Goa trip planning	You give dates, system creates full itinerary

3 Practical Example: AI Recruiter

Scenario

- **Role:** HR Recruiter
- **Task:** Hire a Backend Engineer (remote, 2–4 years experience)
- **Tool:** Company provides an Agentic AI Chatbot

3.1 How the Agentic AI Works

Step 1: Goal Understanding

User provides goal: “I want to hire a backend engineer”

- Remote position
- 2–4 years experience

Step 2: Autonomous Planning

Agent creates a plan:

1. Draft Job Description (JD)
2. Post to best job platforms
3. Monitor applications constantly
4. Adjust strategy if needed
5. Screen candidates
6. Schedule interviews
7. Send offer letters
8. Trigger onboarding

Step 3: Execution with Autonomy

JD Drafting:

- Accesses company documents
- Determines required tech stack (Python, Django)
- Identifies appropriate salary range
- Presents draft for human review

Job Posting:

- Uses LinkedIn & Naukri APIs
- Posts job automatically
- Begins continuous monitoring

Adaptation (Problem Detection):

- Notices only 2 applications received (below expectations)
- Self-identifies problem
- Self-proposes solutions:
 - Broaden JD: Backend → Full Stack
 - Promote job with paid ads on LinkedIn
- Asks for human approval → executes changes

Screening:

- Downloads all resumes using parser tool

- Analyzes against JD
- Categorizes candidates: Strong (2), Partial (3), Weak (3)
- Recommends interviewing strong candidates

Interview Scheduling:

- Checks HR's calendar availability via API
- Proposes Friday schedule
- Sends emails to candidates AND HR

Interview Day:

- Sends reminder
- Provides list of past interview questions

Offer Management:

- Drafts offer letter using company templates
- Sends via email API
- Monitors candidate acceptance

Onboarding:

- Triggers automatically upon acceptance
- Sends welcome email
- Submits IT access requests
- Provisions laptop
- Sets up intro meeting

Key Observation

The agent was given one goal and handled everything autonomously — planning, execution, and adaptation. Human only provided approvals.

4 Key Characteristics of Agentic AI Systems

4.1 1. Autonomy

Definition: Ability to make decisions and take actions independently to achieve a goal without step-by-step human instructions.

Multiple Dimensions of Autonomy:

- **Execution Autonomy:** Automatically executes steps sequentially (JD → Post → Screen → Interview → Offer → Onboard)
- **Decision-Making Autonomy:** Decides which candidates to shortlist and based on what criteria

- **Tool Selection Autonomy:** Decides which tool to use and when (email, calendar, resume parser, HR software)

Controlling Autonomy (Safety Mechanisms)

Control Type	Description	Example
Scope Limits	Limit what tools/actions agent can perform independently	Agent can screen resumes but must ask before rejecting candidates
Human-in-the-Loop	Checkpoints requiring human approval	Must get approval before posting JD or sending offers
Override Controls	Users can stop/-pause/change behavior anytime	“Pause hiring” command stops all activity
Guard Rails & Policies	Hard rules and ethical boundaries	Never schedule interviews on weekends; never use informal language

Why Control is Necessary:

- Agent might roll out incorrect offers without approval
- Agent might show bias in shortlisting (nationality, age)
- Agent might overspend on ads without authorization

4.2 2. Goal-Oriented

Definition: Operates with a persistent objective in mind and continuously directs actions toward achieving that objective, rather than just responding to isolated prompts.

Goal as a Compass: Goals guide the autonomy — without goals, autonomous systems can’t function effectively.

Types of Goals:

- **Independent Goals:** “Hire a backend engineer”
- **Goals with Constraints:** “Hire a backend engineer from India” (location constraint), “Hire remote only” (remote constraint), “Budget \$X” (cost constraint)

Goal Storage (Conceptual JSON Representation):

```
{
  "main_goal": "Hire a backend engineer",
  "constraints": {
    "experience": "2-4 years",
    "remote": true,
    "tech_stack": ["Python", "Django", "AWS"]
  },
}
```

```

"status": "active",
"created_at": "timestamp",
"progress": {
  "jd_drafted": true,
  "job_posted": true,
  "applications": 8,
  "interviews_scheduled": 2,
  "offer_sent": false,
  "onboarding": false
}
}

```

Goal Alteration: Goals can be changed mid-way. Example: “Stop hiring backend engineer. Instead, find a freelancer for this project.”

4.3 3. Planning

Definition: Agent’s ability to break down a high-level goal into a structured sequence of actions and subgoals.

The Two-Step Process

1. **Planning:** Create a structured sequence of subgoals
2. **Execution:** Execute the plan step by step

This process is iterative — if a step fails, the agent returns to planning and creates a new plan.

Three-Step Planning Process

Step 1: Generate Multiple Candidate Plans

- Agent doesn’t create just one plan
- Creates multiple approaches to achieve the same goal
- Example: For hiring a backend engineer:
 - Plan A: Post JD on LinkedIn, GitHub Jobs, Wellfound → Promote → Hire
 - Plan B: Use internal referrals → Hire
 - Plan C: Approach hiring agencies → Hire

Step 2: Evaluate Plans Against Criteria

Criterion		Description
Efficiency		Which plan is faster to execute?
Tool Availability		Does the agent have required tools?
Cost		Which plan fits within budget?
Risk		Which plan has higher failure risk?
Constraint	Align- ment	Which plan best meets constraints (remote, location, etc.)?

Step 3: Select the Best Plan

- Option 1: Human-in-the-loop input (ask supervisor)
- Option 2: Pre-programmed policy based selection

Planning as a Search Problem:

- **Initial State:** Company needs a backend engineer
- **Final State:** Backend engineer hired
- Multiple paths exist between states
- Need to find the most optimal path

4.4 4. Reasoning

Definition: The cognitive process through which an agentic AI system interprets information, draws conclusions, and makes decisions (both while planning and executing).

Analogy: When your phone is stolen:

- **Environment gives information:** Phone is missing
- **Interpret:** Phone has been stolen
- **Draw Conclusion:** Thief might misuse my number
- **Make Decision:** Call Airtel to block the number

Reasoning in Planning Stage

Task	Description
Goal Decomposition	Breaking main goal into sequence of steps
Tool Selection	Determining which tool is needed for each step
Resource Estimation	Estimating time, dependencies, and risks

Reasoning in Execution Stage

Task	Description
Decision Making	Choosing between multiple options (e.g., interview 2 or 3 candidates?)
Human-in-the-Loop	Deciding when to act independently vs. ask for help
Error Handling	Adapting when tools fail (e.g., LinkedIn API down)

4.5 5. Adaptability

Definition: Agent’s ability to modify its plans, strategies, and actions in response to unexpected conditions while staying aligned with the goal.

Scenarios Requiring Adaptation:

Scenario	Example from AI Recruiter
Tool Failures	Calendar API down → Ask human directly about availability
External Feedback	Only 2 applications → Broaden JD and promote job
Goal Changes	Midway: “Stop hiring, find a freelancer instead”

Key Insight: Any agentic AI system will find an alternative path if stuck.

4.6 6. Context Awareness

Definition: Agent’s ability to understand, retain, and utilize relevant information from:

- Ongoing tasks
- Past interactions
- User preferences
- Environmental cues

Types of Context Stored:

Context Type	Examples
Original Goal	“Hire a backend engineer”
Progress Status	JD finalized, posted on LinkedIn, 8 applications received
Conversation History	All agent-human interactions
Environment State	Job posted, 8 applicants, ad budget expiring in 2 days
Tool Responses	“Candidate B has 3 years Django + AWS experience”
User Preferences	Company prefers remote candidates, interview questions in Google Docs
Guard Rails/Policies	“Never send offer letter without approval”

Memory Types

Memory Type	Description	Examples
Short-Term Memory	Current session information	Tool responses, recent decisions, current conversation
Long-Term Memory	Persistent across sessions	High-level goals, past interactions, user preferences, policies

5 Components of Agentic AI Systems

The 5 Core Components

1. *Brain (LLM)*

The backbone that handles heavy lifting:

- Interprets the goal
- Handles planning (breakdown into subgoals)
- Performs reasoning (both planning & execution)
- Handles tool selection
- Generates natural language communication

2. *Orchestrator (The Manager/Project Manager)*

Executes the plan:

- **Task Sequencing:** Which order to execute steps?
- **Conditional Routing:** Based on Step 2 output → go to Step 3 or Step 4?
- **Retry Logic:** Retry if a tool fails
- **Looping/Iteration:** Repeat steps when needed
- **Delegation:** When to involve human vs. LLM vs. tools?

The Orchestrator is like the nervous system and project manager of the agent.

3. *Tools (Hands & Legs)*

Enables interaction with the external world:

- **External Actions:** APIs, database changes, sending emails
- **Knowledge Bases:** RAG-based retrieval of company documents
- **Examples:** LinkedIn API, Resume Parser, Calendar API, Email API, HR Management Software

4. Memory (Storage System)

- **Short-Term Memory:** Current session messages, tool calls, immediate decisions
- **Long-Term Memory:** High-level goals, past interactions, user preferences, policies
- **State Tracking:** What's done? What's pending?

5. Supervisor (Human-in-the-Loop)

Enables human-agent collaboration:

- **Approvals:** High-risk actions (offer letters, paid ads)
- **Guard Rail Enforcement:** Enforcing ethical boundaries
- **Escalations:** Alerting human to edge cases (e.g., non-IIT/NIT candidate with excellent resume)

6 Summary: Identifying an Agentic AI System

To determine if a system is Agentic AI, ask these 6 questions:

#	Characteristic	Question
1	Autonomous	Does it make decisions and act independently?
2	Goal-Oriented	Does it work persistently toward a specific objective?
3	Planning	Does it break down goals into structured sequences?
4	Reasoning	Does it interpret, conclude, and decide?
5	Adaptable	Does it modify plans when unexpected conditions arise?
6	Context-Aware	Does it retain and utilize relevant information?

If all 6 answers are YES → It's an Agentic AI System

7 Key Terms Glossary

Term	Definition
Agentic AI	AI that autonomously works toward achieving user-defined goals
Autonomy	Independent decision-making and action-taking
Goal-Oriented	Persistent focus on achieving a specific objective
Planning	Breaking down goals into structured sequences of actions
Reasoning	Interpreting information, drawing conclusions, making decisions
Adaptability	Modifying plans in response to unexpected conditions
Context Awareness	Understanding and retaining relevant information from tasks & interactions

Term	Definition
Orchestrator	Component that sequences and executes plan steps
Human-in-the-Loop	System checkpoints requiring human approval
Guard Rails	Hard rules and ethical boundaries agents must follow
Short-Term Memory	Current session information
Long-Term Memory	Persistent information across sessions
Tool	External API/service used by agent to perform actions